

CROSS TEST REPORT

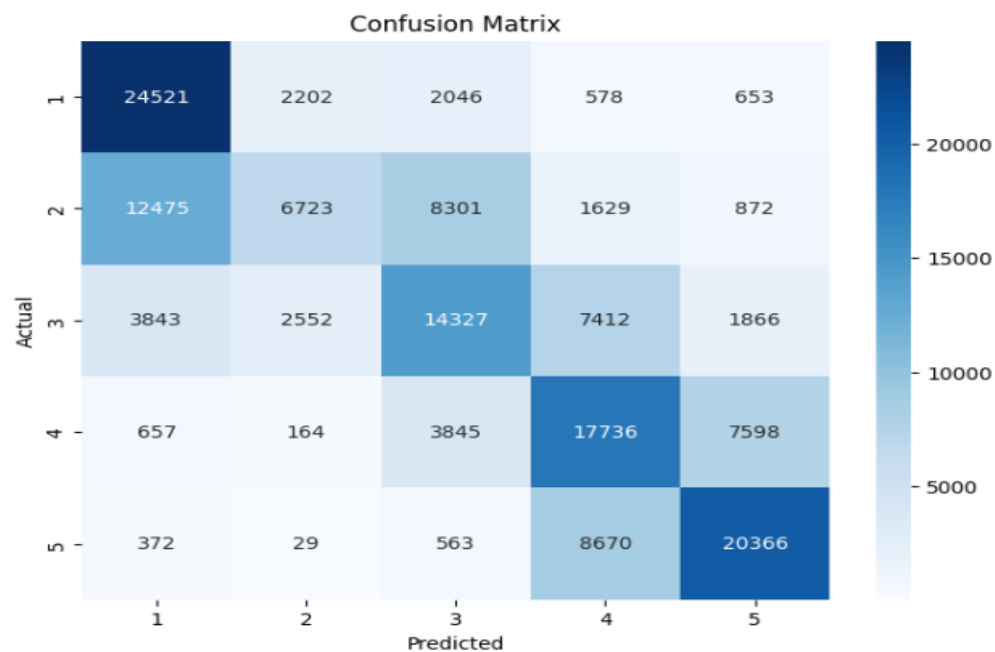
SUMMARY

In this project, two models were trained and evaluated. Model A achieved an accuracy of 64%, whereas Model B achieved 54% accuracy.

- Model A (Imbalanced training) tested on balanced data → 56% accuracy
- Model B (Balanced training) tested on imbalanced data → 53% accuracy
- Model A performs slightly better than Model B in cross testing.
- Accuracy difference is small but still in favor of Model A.

Model A tested using balanced dataset:

Accuracy: 0.55782				
Classification Report:				
	precision	recall	f1-score	support
1	0.59	0.82	0.68	30000
2	0.58	0.22	0.32	30000
3	0.49	0.48	0.48	30000
4	0.49	0.59	0.54	30000
5	0.65	0.68	0.66	30000
accuracy			0.56	150000
macro avg	0.56	0.56	0.54	150000
weighted avg	0.56	0.56	0.54	150000

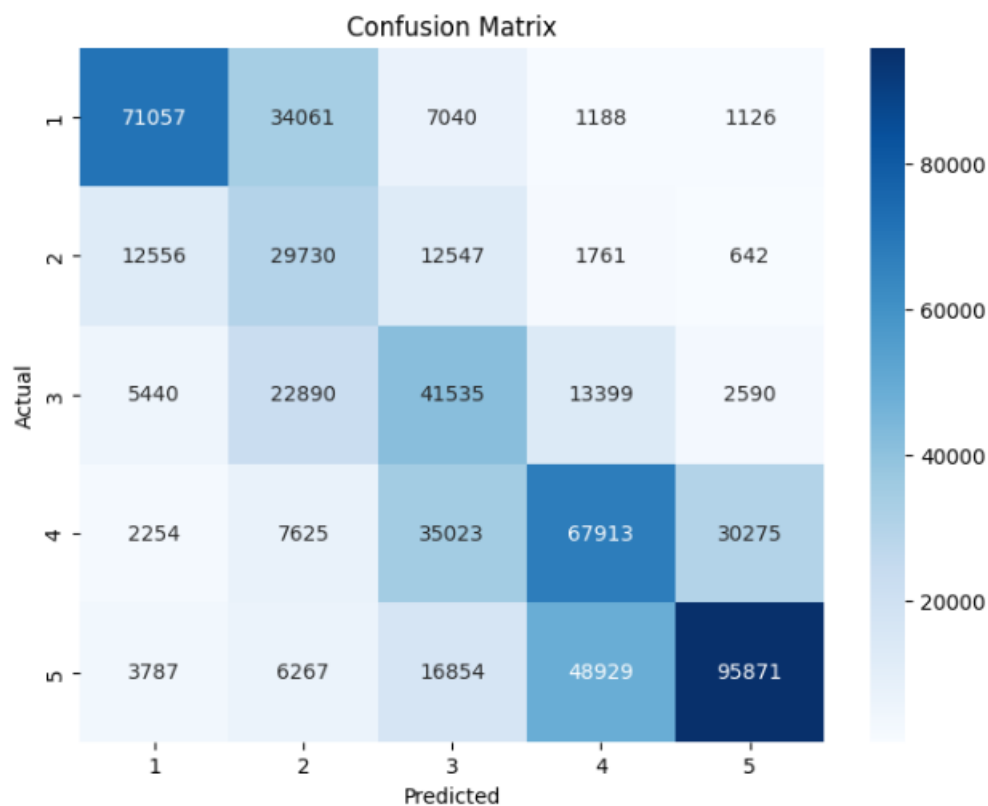


Model B tested using the imbalanced dataset:

Accuracy: 0.5348137535816619

Classification Report:

	precision	recall	f1-score	support
1	0.75	0.62	0.68	114472
2	0.30	0.52	0.38	57236
3	0.37	0.48	0.42	85854
4	0.51	0.47	0.49	143090
5	0.73	0.56	0.63	171708
accuracy			0.53	572360
macro avg	0.53	0.53	0.52	572360
weighted avg	0.58	0.53	0.55	572360



OBSERVATIONS:

- ☐ Training on imbalanced data makes Model B biased toward majority classes.
- ☐ Model B performs poorly when the data distribution changes.
- ☐ Model A was trained on balanced data and learned to recognize all classes.
- ☐ Because of that, Model A generalizes slightly better across unseen distributions.

- ☐ Data balancing clearly improves model robustness and fairness.

RECOMMENDATION: choosing the model

- ☐ Deploy Model A because it generalizes better.
- ☐ It handles both balanced and imbalanced inputs more reliably.
- ☐ Even with small accuracy improvements, Model A is more stable and less biased.
- ☐ Use Model B only as a baseline or fallback model if needed.